EE5121 - Assignment 1

Rohith R - EP21B030

Contents

Question 1 **2**
- ℓ_0-norm sparse solution . . . 2
- ℓ_1-norm proxy . . . 2
- ℓ_1 as a linear program . . . 2
- Application ($Ax = b$ is feasible) . . . 3
- Application ($Ax = b$ is infeasible) . . . 3

Question 2 **4**
- Rank Minimization . . . 4
- Nuclear Norm Minimization . . . 5
- Multi-experiment system identification . . . 5

Question 3 **7**
- Feasibility via 0-1 formulation . . . 7
- SDP relaxation - vector K-coloring . . . 7
- Application in Physics: Frequency Assignment in Radio Astronomy . 8

Question 1

a). ℓ_0-norm sparse solution

No, the given optimization problem is not convex. The constraint set ($Ax = b$) is a convex set but the objective function ($\|x\|_0$) is not a convex function.

Counter-example: Let $x = (1, 0)$ and $y = (0, 1)$. Then $\|x\|_0 = 1$, $\|y\|_0 = 1$, but for their midpoint $\frac{1}{2}(x+y) = (0.5, 0.5)$, we have $\|\frac{1}{2}(x+y)\|_0 = 2 > \frac{1}{2}\|x\|_0 + \frac{1}{2}\|y\|_0 = 1$. Hence, $\|\cdot\|_0$ is not convex.

b). ℓ_1-norm proxy

Yes, the given optimization problem is convex as the constraint set ($Ax = b$) is a convex set and the objective function $\|x\|_1$ is a convex function.

Proof:
Proving the objective function is convex
Let $x, y \in \mathbb{R}^n$ and let $\theta \in [0, 1]$. For each component i we have

$$\left|\theta x_i + (1-\theta)y_i\right| \leq \left|\theta x_i\right| + \left|(1-\theta)y_i\right| = \theta|x_i| + (1-\theta)|y_i|,$$

summing over $i = 1, \ldots, n$ yields

$$\|\theta x + (1-\theta)y\|_1 = \sum_{i=1}^{n} \left|\theta x_i + (1-\theta)y_i\right| \leq \sum_{i=1}^{n} \left(\theta|x_i| + (1-\theta)|y_i|\right) = \theta\|x\|_1 + (1-\theta)\|y\|_1.$$

Thus $\|\cdot\|_1$ is convex.

Proving the constraint set is convex
Let $x, y \in \mathbb{R}^n$ s.t. $Ax = b$ and $Ay = b$, let $\theta \in [0, 1]$.

$$A(\theta x + (1-\theta)y) = \theta Ax + (1-\theta)Ay = \theta b + (1-\theta)b = b$$

Therefore $\{x : Ax = b, x \in \mathbb{R}^n\}$ is convex (Infact, it's an affine set).

c). ℓ_1 as a linear program

Original $\implies$ LP
Take any feasible x for the original problem i.e. $Ax = b$. Let $u_i := |x_i|$ for each i then,

$$-u_i \leq x_i \leq u_i \text{ and } u_i \geq 0$$
$$\mathbf{1}^T u = \sum_i u_i = \sum_i |x_i| = \|x\|_1$$

Thus (x, u) is feasible for the LP, and the objective of the LP equals the objective of the original problem.

LP $\implies$ Original Take any feasible (x, u) for the LP i.e. $-u_i \leq x_i \leq u_i \implies |x_i| \leq u_i \forall i$ and $Ax = b$ so,

$$\|x\|_1 = \sum_i |x_i| \leq \sum_i u_i = \mathbf{1}^T u$$

Thus at optimality, $\mathbf{1}^T u = \|x\|_1$, recovering the original problem.

d). Application ($Ax = b$ is feasible)

Framing the problem

We wish to synthesize the target reflectance $b \in \mathbb{R}^3$ using the pigment responses in the columns of $A \in \mathbb{R}^{3\times 3}$. Let $x \in \mathbb{R}^3$ denote the mixture weights of pigments P_1, P_2, P_3. The fewest-pigments problem can be posed as

$$\min_x \ \|x\|_0 \quad \text{s.t.} \quad Ax = b, \ \ x \geq 0,$$

where $\|x\|_0$ counts the number of nonzero entries of x (the number of pigments used).

Since the ℓ_0 objective is nonconvex, we replace it by its convex surrogate, the ℓ_1 norm. This yields the *sparse recovery* formulation

$$\min_x \ \|x\|_1 \quad \text{s.t.} \quad Ax = b, \ \ x \geq 0,$$

which can be solved efficiently as a linear program as seen above.

Solving the problem

Code: Click here
Optimizer x^*: $[0., 0.6, 0.2]$
Support (1-based indices): $\{2, 3\}$

e). Application ($Ax = b$ is infeasible)

Framing the problem

Since the system $Ax = b$ may be infeasible due to model error or measurement noise, we replace the exact equality constraint with a data-fitting penalty. We formulate the convex program

$$\min_{x\geq 0} \ \mathbf{1}^\top x \ + \ \gamma \, \|Ax - b\|_2^2,$$

where the first term $\mathbf{1}^\top x = \|x\|_1$ promotes sparsity (few pigments used) and the second term $\|Ax - b\|_2^2$ penalizes model mismatch. The parameter $\gamma > 0$ balances sparsity against fidelity to the target reflectance b. **Note:** This is in some sense the same as the MAP estimate of x given b where $b = Ax + \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$ subject to laplacian prior (which promotes sparsity).

Solving the problem

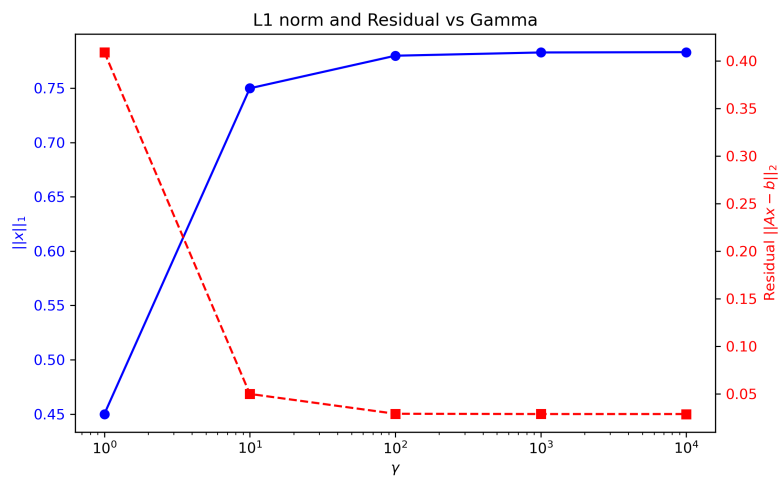

Figure 1: Plot showing the L1-norm of the optimizer and the residual vs γ

Since all solutions for γ between 10 and 10,000 have a support size of 1, and further increasing γ leads to a support size of 3, we report the results at $\gamma = 10{,}000$.

Code: Click here
Optimizer x^*: $[0., 0.599935, 0.183365]$
Support (1-based indices): $\{2, 3\}$
Residual: 0.02886755

Question 2

a). Rank Minimization

No, the given optimization problem is not convex. The constraint set ($AX = B$) is a convex set but the objective function ($\text{rank}(X)$) is not a convex function.

Counter-example: Let

$$X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then $\text{rank}(X_1) = \text{rank}(X_2) = 1$. However,

$$\frac{1}{2}X_1 + \frac{1}{2}X_2 = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

has rank 2. Therefore

$$\operatorname{rank}\left(\tfrac{1}{2}X_1 + \tfrac{1}{2}X_2\right) = 2 \not\leq \tfrac{1}{2}\cdot 1 + \tfrac{1}{2}\cdot 1 = 1,$$

which violates convexity. Hence, rank minimization is a nonconvex problem.

b). Nuclear Norm Minimization

Proving the convexity of Nuclear norm
The nuclear norm is defined by

$$\|X\|_* = \sup_{\|Y\|_2 \leq 1} \langle X, Y \rangle,$$

where $\langle X, Y \rangle = \operatorname{Tr}(X^\top Y)$.

Proof:
For each fixed Y with $\|Y\|_2 \leq 1$ the function $\langle X, Y \rangle$ is linear (hence convex). The nuclear norm is the pointwise supremum of this family of linear functions. Since the supremum of an arbitrary family of convex functions is convex, it follows that $\|\cdot\|_*$ is convex.

c). Multi-experiment system identification

Solving: Without noise
We solve the matrix equation

$$AX = B$$

by minimizing the nuclear norm of X using CVXPY.

Code: Click here
Solver status: optimal
Optimal nuclear norm: 7.9281
Optimal solution X^*:

$$X^* \approx \begin{bmatrix} 1 & -1 & 0 & 0 & 1 \\ 2 & 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 \\ 2 & -2 & 0 & 0 & 2 \\ -2 & 0 & -1 & 1 & 0 \end{bmatrix}.$$

Note: Small numerical noise (order 10^{-8}) has been rounded to zero for clarity.

The toeplitz structure of A is due to the convolution operation which uniquely defines the output of a LTI system given an input through impulse response.

Solving: With noise
We consider the matrix equation

$$AX \approx B_{\text{noisy}}$$

where $B_{\text{noisy}} = B + \sqrt{0.1}N$ and $N \sim \mathcal{N}(0, I)$. The goal is to recover a matrix X with minimal nuclear norm while fitting the noisy data. Like before, this is formulated as the convex optimization problem:

$$\min_{X \in \mathbb{R}^{5\times 5}} \lambda \|X\|_* + \frac{1}{2}\|AX - B_{\text{noisy}}\|_F^2$$

where $\|X\|_*$ denotes the nuclear norm of X and $\lambda > 0$ controls the trade-off between low-rank structure and data fidelity.
Figure 2 shows how the nuclear norm of the recovered matrix X^* and the residual $\|AX - B_{\text{noisy}}\|_F$ vary with the regularization parameter λ.

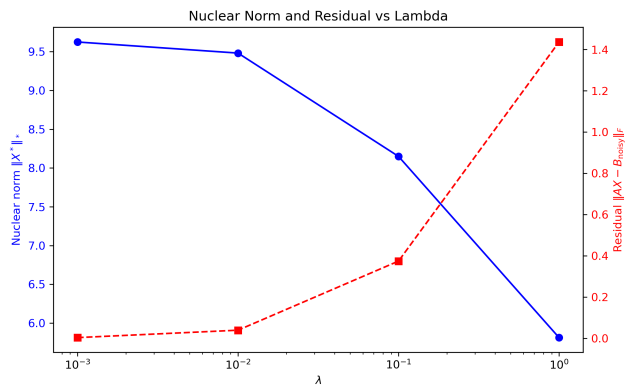

Figure 2: Nuclear norm and residual vs regularization parameter λ. Smaller λ emphasizes data fitting (smaller residual), while larger λ encourages low-rank solutions (smaller nuclear norm).

From the plot, I select $\lambda = 0.1$ as a reasonable trade-off between data fidelity and low-rank structure. The corresponding results are:

Code: Click here
Solver status: optimal
Optimal nuclear norm ≈ 8.1498
Residual ≈ 0.3748
Optimal solution X^***:**

$$X^* \approx \begin{bmatrix} 1.1041 & -0.9969 & 0.3421 & -0.0502 & 0.9824 \\ 1.9297 & 0.3898 & 0.8883 & -1.1658 & -0.4822 \\ 0.8937 & 0.6721 & 0.5185 & -0.9338 & -0.9070 \\ 1.7428 & -1.7806 & 0.0457 & 0.1274 & 2.0998 \\ -1.8097 & -0.0218 & -1.2021 & 1.0649 & 0.4559 \end{bmatrix}.$$

Question 3

a). Feasibility via 0-1 formulation

Is the constraint set convex?
No, take G to be a graph with only a single node, i.e. $V = \{1\}$ and $E = \phi$, and take $K = 2$. Now one feasible point in the constraint set is $x^{(1)} = (x^1_{11}, x^1_{12}) = (1, 0)$ and another feasible point in the set is $x^{(2)} = (x^2_{11}, x^2_{12}) = (0, 1)$, now, their midpoint is

$$\tfrac{1}{2}(x^{(1)} + x^{(2)}) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right),$$

which does not satisfy the binary constraint. Thus the constraint set with $x_{vk} \in \{0, 1\}$ is not convex.

After dropping the binary requirement?
Yes, after relaxing the binary constraint and only requiring $0 \leq x_{vk} \leq 1$, the constraint set is convex.
The constraints are as follows.

$$\sum_{k=1}^{K} x_{vk} = 1, \qquad x_{uk} + x_{vk} \leq 1, \qquad 0 \leq x_{vk} \leq 1,$$

which are all linear equalities or inequalities. Therefore, the feasible set is the intersection of affine and halfspace constraints, i.e., a convex polytope. Hence the relaxed set is convex.

b). SDP relaxation - vector K-coloring

Meaning of feasibility/infeasibility
If it's feasible: Then we are able to find a PSD matrix which can always be represented as a Gram matrix, and when the constraint $G_{uv} \leq -\dfrac{1}{K-1}$ $(\forall (u, v) \in E)$ is satisfied then, we can assign to each vertex an unit vector such that the inner product between any two of them is $\leq -\dfrac{1}{K-1}$ and **we have a K-coloring** of the graph G.

If it's infeasible: Then we have $G_{uv} > -\dfrac{1}{K-1}$ (for some $(u, v) \in E$) and **the graph contains a K-clique**.

Equivalence to minimizing edge-wise inner-product
Consider the feasibility problem (F_K):

$$\begin{aligned} &\text{find } G \in \mathbb{S}^n \\ &\text{s.t. } G \succeq 0, \quad G_{vv} = 1 \ \ \forall v, \quad G_{uv} \leq -\tfrac{1}{K-1} \ \ \forall (u,v) \in E. \end{aligned}$$

Now define the optimization problem (P):

$$\begin{aligned} &\min_{G,\rho} \ \rho \\ &\text{s.t. } G \succeq 0, \quad G_{vv} = 1 \ \ \forall v, \quad G_{uv} \leq \rho \ \ \forall (u,v) \in E. \end{aligned}$$

- If (F_K) is feasible, then there exists G_0 with $G_{0,uv} \leq -\frac{1}{K-1}$ on all edges. Setting $\rho = -\frac{1}{K-1}$ yields a feasible point for (P), so the optimal value ρ^* satisfies $\rho^* \leq -\frac{1}{K-1}$.
- Conversely, if $\rho^* \leq -\frac{1}{K-1}$ for an optimal solution (G^*, ρ^*) of (P), then G^* also satisfies $G^*_{uv} \leq -\frac{1}{K-1}$, hence is feasible for (F_K).

Therefore,

$$(F_K) \text{ is feasible} \quad \Longleftrightarrow \quad \rho^* \leq -\tfrac{1}{K-1}.$$

Chromatic Number of the graph
From the SDP above, if the optimal value satisfies $\rho^\star \leq -\frac{1}{K-1}$, the graph admits a vector K-coloring, hence $\chi(G) \leq K$. If instead $\rho^\star > -\frac{1}{K-1}$, the graph is not K-colorable, so $\chi(G) > K$.

c). Application in Physics: Frequency Assignment in Radio Astronomy

In large radio telescope arrays (such as the VLA or SKA), different antennas or detectors must operate simultaneously. However, nearby antennas or those with overlapping fields of view cannot use the same frequency band due to interference. The task is to assign *frequency channels* so that interfering antennas do not share the same frequency.

This situation maps naturally to a graph coloring problem:

- Vertices V correspond to antennas (or detectors).
- An edge $(u,v) \in E$ indicates that antennas u and v interfere and therefore cannot use the same frequency.
- Colors correspond to frequency channels.

The vector K-coloring formulation leads to the following feasibility SDP:

$$\begin{aligned} &\text{find } G \in \mathbb{S}^n \\ &\text{s.t. } G \succeq 0, \\ &\quad G_{vv} = 1 \quad \forall v \in V, \\ &\quad G_{uv} \leq -\tfrac{1}{K-1} \quad \forall (u,v) \in E. \end{aligned}$$

Here:

- n = number of antennas in the array,
- each unit vector u_v represents the frequency group assigned to antenna v,
- the constraint $G_{uv} \leq -\frac{1}{K-1}$ enforces that interfering antennas must be assigned to different groups.

Interpretation: A feasible solution for this SDP with parameter K means that it is possible to operate the telescope array using only K frequency bands, while ensuring that no two interfering antennas use the same band.